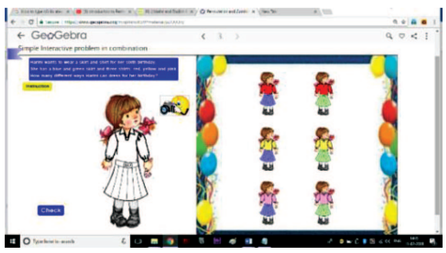
Summary

In this chapter, we acquired the knowledge of

- Factorial of a natural number n is the product of the first n natural numbers.
- $n! = n(n-1)!$, for any integer $n \geq 1$.
- The number of ways of arranging n unlike objects is $n!$.
- The number of distinct permutations of r objects which can be made from n distinct objects is
$$^nP_r = \frac{n!}{(n-r)!} = n(n-1)(n-2)\cdots(n-r+1).$$
- The number of permutations of n objects taken all at a time where P_1 objects one of first kind, P_2 objects one of second kind, $\cdots$ P_k objects one of the kth kind and the rest, if any are all different and is given by
$$\frac{n!}{P_1!P_2!\cdots P_k!}.$$
- Order matters for a permutation where as order does not matter for a combination.
- The number of combinations of n different objects taken r at a time denoted by ^nC_r is given by
$$^nC_r = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$$

ICT CORNER-4(a)

Expected Outcome $\Rightarrow$



Step–1: Open the Browser and type the URL Link given below (or) Scan the QR Code. GeoGebra worksheet Permutations and Combination will appear. You can select the worksheet you want to open, for example open Combination Game

Step–2: Press Instruction button and read how to play. After reading once again press Instruction button and press Start the game.

Step–3: Find the number of combination to dress up the girl with 3 Shirts and 2 Skirts. After selecting each pair, click on Camera picture, so that the selected will appear in right space. After completing the selection press check button to see the result.